3000747 Healthcare Innovation and Entrepreneurship

Building Your First AI-Powered Clinical Prototype: A Hands-On Worksheet

Wednesday, March 25th, 2026

Outline

Part	Topic	Duration	Core Competency
Part A	Technical Infrastructure	15 Mins	API Security, MCP Architecture & Resource Management
Part B	Vibe Coding & UI Design	30 Mins	Natural Language Prototyping & Interface Transparency
Part C	Prompt Engineering	45 Mins	Few-Shot Logic & Algorithmic Bias Mitigation
Part D	Clinical Validation	30 Mins	Accuracy Auditing & Legal Liability Frameworks

SESSION OVERVIEW & LEARNING OBJECTIVES

In the modern healthcare landscape, the ability to build and control AI is becoming just as important as clinical expertise. We are moving away from "black box" systems—where hidden algorithms dictate care—toward a new era of **clinician-led prototyping**.

By using **"Vibe Coding"** (using natural language to generate code), you are no longer just a passive user of technology; you are becoming a **Clinical AI Architect**. This shift allows you to directly solve the inefficiencies and documentation burdens you face every day, ensuring that AI tools are tailored to the actual realities of the bedside.

Upon completion of this session, you will be able to:

- **Move from Theory to Action:** Transition from simply "knowing" about AI to "doing" through rapid prototyping and iterative design.
- **Master the Prompting Hierarchy:** Evaluate the utility of "Zero-shot" versus "Few-shot" techniques to ensure clinical accuracy.
- **Build a Working Tool:** Construct a functional **Medical SOAP Note Assistant** using a modern technical stack (Lovable.dev + Gemini API).
- **Critique with Confidence:** Navigate the ethical implications of AI, specifically focusing on how to manage algorithmic bias, "hallucinations," and the legal nuances of liability.

PART A: TECHNICAL INFRASTRUCTURE & API SECURITY (SETUP)

Technical literacy has become a "foundational competency" for the modern physician. For healthcare leaders, mastering these technical underpinnings is essential to avoid wasting institutional resources and to prevent the safety hazards of misconfigured systems.

While this prototype uses a simple API connection to get you started, "production-grade" clinical applications are becoming more sophisticated. Real-world tools increasingly rely on the **Model Context Protocol (MCP)** and robust database integrations, such as **Supabase**, to securely manage complex patient data and handle professional authentication.

Step-by-Step Configuration

Lovable.dev Account Setup:

- Navigate to Lovable.dev and sign up.
- Access your dashboard, which serves as your environment for converting natural language into functional React applications.

Google AI Studio Integration:

- Navigate to aistudio.google.com.
- Generate a new API key under the "API Keys" menu.

Model Selection: We will utilize **Gemini 3 Flash** for rapid, responsive interface builds. For Part C, we will transition to **Gemini 3.1 Pro** for its advanced reasoning, software engineering capabilities, and more grounded responses. Note that Gemini 3 Pro Preview was shut down in March 2026; ensure you are using the stable 3.1 Pro release.

Environment Variable Isolation: Never hardcode keys into your prompt text. Use Lovable's "Connectors" or environment variables to prevent key leakage.

Strict Anonymization: Generative models are susceptible to re-identification attacks. Remove all PHI (names, exact dates, ID numbers) before processing clinical text in a non-HIPAA-compliant testing environment.

Governance & Rate Limits: Monitor your "Usage and Limits" in Google AI Studio. Exceeding your 429 rate limits during a clinical simulation can lead to service interruptions that compromise care continuity.

Questions for Reflection: Reflecting on standard data privacy concerns in digital health, how would you propose a mitigation strategy—such as using environment variables or a dedicated **Model Context Protocol (MCP)** server—to prevent a physician from being held legally liable for a data breach caused by improper API key storage?

PART B: VIBE CODING & UI DESIGN (THE SOAP ASSISTANT)

"Vibe Coding" represents the democratization of software development, where natural language replaces manual syntax. This allows the medical professional to dictate the "vibe"—the form, function, and logic—of a tool without the technical barrier of traditional coding.

Activity 1: The Zero-Shot UI (Addressing the Black Box)

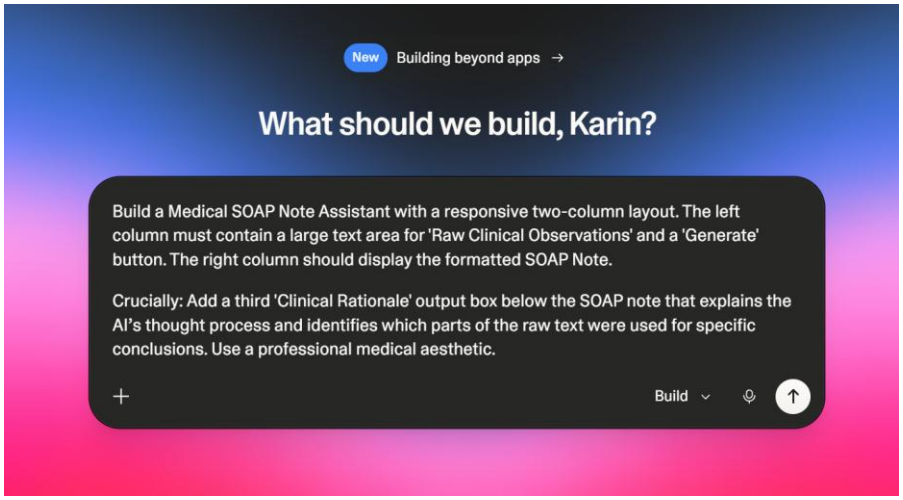
In **Lovable.dev**, enter the following prompt to build your interface. Note that including a "Rationale" box specifically addresses the "Black Box" transparency problem by showing how the AI reaches its conclusions.

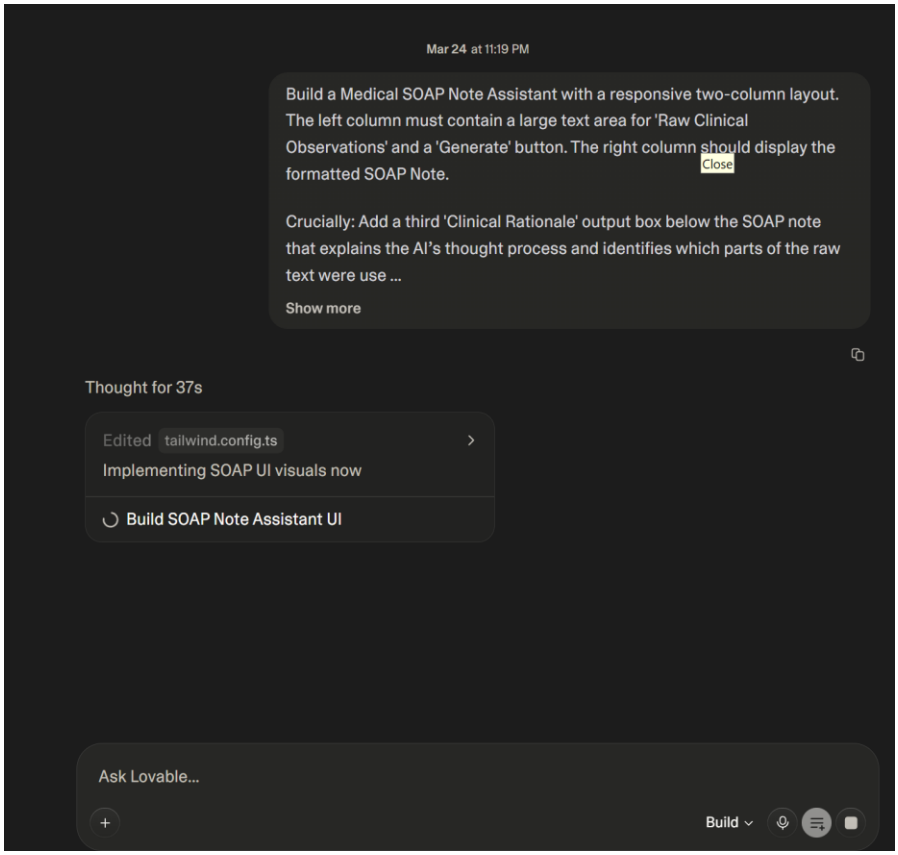
> Build a **Medical SOAP Note Assistant** with a responsive two-column layout. The left column must contain a large text area for **'Raw Clinical Observations'** and a **'Generate'** button. The right column should display the **formatted SOAP Note**. For the 'Raw Clinical Observations' part, a user should be able to configure the AI's system prompt.
> **Crucially:** Add a third **'Clinical Rationale'** output box below the SOAP note that explains the AI's thought process and identifies which parts of the raw text were used for specific conclusions. Use a professional medical aesthetic.

Questions for Reflection

- How does providing a "Rationale" box change your trust in the AI's output compared to a simple summary?

- How else would you improve this design to better serve a high-pressure clinical environment? Consider how the app could evolve to handle **multi-modal data** (like voice or images), reduce **cognitive load** for the physician, or more effectively prevent **medical errors**?


New
Building beyond apps →
What should we build, Karin?
Build a Medical SOAP Note Assistant with a responsive two-column layout. The left column must contain a large text area for 'Raw Clinical Observations' and a 'Generate' button. The right column should display the formatted SOAP Note.
Crucially: Add a third 'Clinical Rationale' output box below the SOAP note that explains the AI's thought process and identifies which parts of the raw text were used for specific conclusions. Use a professional medical aesthetic.
Build


Mar 24 at 11:19 PM
Build a Medical SOAP Note Assistant with a responsive two-column layout. The left column must contain a large text area for 'Raw Clinical Observations' and a 'Generate' button. The right column should display the formatted SOAP Note.
Close
Crucially: Add a third 'Clinical Rationale' output box below the SOAP note that explains the AI's thought process and identifies which parts of the raw text were use ...
Show more
Thought for 37s
Edited tailwind.config.ts
Implementing SOAP UI visuals now
Build SOAP Note Assistant UI
Ask Lovable...
Build

Close

Raw Clinical Observations Load example

```
Pt 45yo male, chest pain x2 days, worse with exertion. BP 140/90.
```

Generate SOAP Note

Visual edits Build

Close

FORMATTED SOAP NOTE

S — SUBJECTIVE

54-year-old male presents with acute onset chest pain, 2 hours duration. Describes squeezing sensation in the center of the chest with radiation to the left arm. Pain rated 7/10. PMH significant for hypertension and type 2 diabetes mellitus. Current medications include metformin 500mg BID and lisinopril 10mg daily.

O — OBJECTIVE

Vitals: BP 158/94 mmHg (elevated), HR 92 bpm, Temp 98.6°F (afebrile), SpO2 97% on RA.
General: Anxious, diaphoretic.
ECG: ST-segment depression in leads V4-V6.
Labs: Troponin levels pending.

A — ASSESSMENT

Acute coronary syndrome (ACS), likely unstable angina vs NSTEMI given the clinical presentation of typical chest pain with ST-segment depression. Elevated blood pressure may be secondary to pain and anxiety. Troponin results will help differentiate.

P — PLAN

1. Obtain serial troponins q6h
2. Administer aspirin 325mg stat, nitroglycerin SL PRN
3. Start heparin drip per ACS protocol
4. Cardiology consult for possible cardiac catheterization
5. Continue home medications; hold metformin pending renal function
6. Continuous telemetry monitoring
7. NPO pending potential intervention

Clinical Rationale

`Subjective → Assessment`
Typical anginal chest pain pattern identified
Source: "squeezing sensation in the center of his chest, radiating to his left arm"

`Objective → Assessment`
ECG findings support ischemic etiology
Source: "ECG shows ST-segment depression in leads V4-V6"

`Objective → Plan`
Elevated BP warrants monitoring but likely reactive
Source: "BP 158/94"

`Subjective → Plan`
Metformin held due to potential contrast exposure
Source: "Currently on metformin 500mg BID"

`Objective → Assessment`
Diaphoresis as autonomic stress marker reinforces ACS concern
Source: "Patient appears anxious and diaphoretic"

Activity 2: Iterative Refinement

Objective: Transform your basic prototype into a high-utility clinical assistant by layering features that address safety, accessibility, and diagnostic depth.

The following table outlines the 10 features you will implement over the next two weeks to transform your prototype into a professional medical tool.

#	Feature	Prompting Strategy	Clinical Rationale & Impact
1	**Human-in-the-Loop "Edit Mode"**	*"Add an 'Edit' toggle to the generated SOAP note. When active, turn the text into an editable field."*	Reinforces that the physician, not the AI, provides the final clinical signature .
2	**Visual Confidence Scoring**	*"Highlight sentences in yellow if the AI has low confidence in that specific clinical inference."*	Directs limited physician attention toward potential hallucinations or errors.
3	**Patient-Friendly Translation**	*"Add a 'Patient Mode' button to generate a version in plain, non-medical language (5th-grade level)."*	Bridges the digital divide and improves patient health literacy and engagement .
4	**"Hard Stop" Safety Checklist**	*"Disable 'Copy to Clipboard' until I check a box: 'I have verified all dosages, allergies, and findings.'"*	Mitigates "Automation Bias" by forcing a cognitive pause before data entry .
5	**High-Contrast UI Mode**	*"Add an 'Accessibility' toggle to increase font size to 18px and switch to high-contrast Black/White."*	Ensures the tool remains inclusive for senior consultants and those with visual impairments.
6	**Differential Diagnosis (DDx) Expander**	*"Include an 'Explore Alternatives' button to list three other possible diagnoses based on raw data."*	Encourages "thinking longer" to prevent premature closure in clinical reasoning .
7	**"Red Flag" Emergency Detection**	*"Automatically highlight in bold red any life-threatening symptoms (e.g., chest pain) in the raw notes."*	Acts as a real-time safety net to ensure critical findings are never overlooked.
8	**One-Click Specialist Referral**	*"Add a dropdown for 'Specialty Referral' and a button to re-format the note into a formal referral letter."*	Reduces administrative burnout by automating repetitive clerical and formatting tasks .
9	**Multi-Modal Support**	*"Add a file upload zone so the AI can analyze an image (lab report or rash photo) alongside text."*	Enables a holistic synthesis of visual and textual clinical evidence .
10	**Voice-to-Note "Ambient" Interface**	*"Integrate a microphone icon that uses Voice Mode to dictate observations into the 'Raw Notes' box."*	Encourages natural patient-doctor eye contact during the clinical consultation .

PART C: PROMPT ENGINEERING & CLINICAL LOGIC

The accuracy of a clinical assistant depends on the hierarchy of prompting. While "Zero-shot" (simple commands) is fast, "Few-shot" prompting (providing examples) is essential for mitigating systemic bias and preventing "hallucinations"—the generation of false medical data.

Implementation Task: The Few-Shot System Prompt

Input the following logic into the **Lovable "Instructions"** panel or the **Gemini System Instructions** to link your UI to the model.

System Prompt Template:

Role: You are an expert Physician-Developer specializing in Digital Health . **Task:** Convert raw clinical notes into a structured SOAP format while maintaining high diagnostic fidelity . **Constraints:** Do not speculate on diagnoses not supported by the text. If data is missing (e.g., Objective vitals), list them as "Not Recorded" .

Few-Shot Examples :

1. **Example 1 (Standard):**
   1.1. *Input:* Pt 45yo male, chest pain x2 days, worse with exertion. BP 140/90.
   1.2. *Output:* **S:** 45yo M reports exertional chest pain for 48 hours. **O:** BP 140/90. **A:** Chest pain, suspicious for angina. **P:** Order EKG and Troponin .
2. **Example 2 (Demographic Diversity/Bias Mitigation):**
   2.1. *Input:* Low-income female pt, 30yo, presents with chronic fatigue. History of infrequent visits due to cost.
   2.2. *Output:* **S:** 30yo F reports persistent fatigue. **A:** Fatigue, etiology unclear. **P:** Complete Metabolic Panel (CMP). *Note: Ensure clinical severity is prioritized over historical lack of testing*.

Activity: Prompt Refinement & Iteration

Once the initial prompt is set, you must "tune" the AI's tone and output format.

- **Task:** Tell the AI to adjust its tone. Example: *"Make the Assessment section more detailed for a Specialist audience"* or *"Ensure the Plan section uses bullet points for clarity"*.
- **Validation:** Observe how changing a single sentence in the system prompt shifts the entire structure of the generated SOAP note.

Activity: Prompt Refinement & Iteration

Loop 1: The "Hallucination" Check (Data Integrity)

- **The Task:** Provide intentionally vague or contradictory notes: *"Patient says they have a 'raging fever,' but forehead feels cool to the touch; the digital thermometer reads 36.5°C"*.
- **Evaluation:** Does the AI correctly prioritize the **objective data** (36.5°C) in the Assessment, or does it falsely claim "Patient has a fever" because the patient said so?
- **Fix:** If it fails, add: *"Constraint: Always prioritize objective vitals over subjective patient statements in the Assessment section"*.

Loop 2: The "Negative Findings" Check (Omission)

- **The Task:** Input a note that specifically mentions what the patient *doesn't* have: *"Pt denies chest pain, denies shortness of breath, denies dizziness."*
- **Evaluation:** Does the AI accidentally list "Chest pain" in the SOAP note because the words appeared in the text?
- **Fix:** If it fails, add: *"Constraint: Distinctly categorize 'Negative Findings' or 'Denials' to avoid false-positive symptoms"*.

Loop 3: The "Acronym & Jargon" Check (Clarity)

- **The Task:** Use heavy clinical shorthand: *"65yo M c/o SOB and DOE, hx of CHF and AFib, currently on NOACs."*
- **Evaluation:** Does the AI correctly expand these into "Shortness of Breath," "Dyspnea on Exertion," "Congestive Heart Failure," and "Non-vitamin K Antagonist Oral Coagulants"?
- **Fix:** If it fails, add: *"Example: Input: SOB/DOE. Output: Subjective: Shortness of breath and dyspnea on exertion"*.

Questions for Reflection:

How does providing "Few-shot" examples involving diverse patient demographics help mitigate the specific risk of "cost-proxy" bias, where AI might falsely prioritize patients based on historical spending rather than clinical severity?

PART D: SIMULATED ENCOUNTER & CLINICAL VALIDATION

Validation is the bridge between a "cool demo" and a safe medical tool. While AI can significantly boost a clinician's confidence in handling documentation, that confidence must be balanced by a rigorous audit of the AI’s logic and its potential for "hallucination".

The Accuracy Audit: Standard Test Cases

Use the following scenarios to test if your "Clinical Brain" (the prompt you built in Part C) is functioning correctly.

Test Case (Input)	Expected AI Behavior (Output)	Why This Matters
The Contradiction: Pt says "I have a fever," but later mentions "My temperature was 36.6°C this morning."	The **Objective** section must list 36.6°C. The **Assessment** should note "Subjective fever, objectively afebrile."	Tests if the AI prioritizes hard data over patient narrative .
The Missing Info: "Patient presents with a cough and sore throat." (No mention of smoking or allergies).	The AI should **not** invent a smoking status. It should list Smoking/Allergies as "Not Recorded."	Tests for **Hallucinations** (making up data that wasn't there).
The Dosage Trap: "Start Lisinopril 10mg once daily."	The **Plan** must state exactly "10mg." It should not suggest a different dose or frequency.	LLMs are prone to "averaging" medical knowledge; precision is life-critical .

Activity: Design Your Own "Edge-Case" Scenario

Now, go beyond the basics. Use an LLM (ChatGPT/Gemini) to help you generate a "messy" patient transcript to see where your app breaks .

- **Step 1 (Generate):** Ask an AI: "Generate a realistic but disorganized 10-sentence transcript of a patient who is an unreliable historian, contradicts themselves, and uses heavy slang."
- **Step 2 (Test):** Paste that transcript into your app's "Raw Notes" box.
- **Step 3 (Audit):** Use the **Rationale Box** to check:
    - Did the AI catch the contradictions?
    - Did it ignore the slang and stick to clinical facts?
    - Did it "invent" any diagnoses to make sense of the mess?

What if the result isn't as expected?

If the AI fails (e.g., it misses a symptom or hallucinates a dose), return to your **System Prompt** in Part C :

1. Add a specific **Constraint** (e.g., "If data is contradictory, flag both in the Assessment").
2. Add a **Few-shot Example** showing the AI how to handle that exact mess.

Questions for Reflection

- **The Liability Gap:** If your AI suggests an incorrect medication dosage that leads to a patient injury, who is legally responsible? Is it the developer, the hospital, or you—the physician who signed off on the note?

- **Automation Bias:** If the UI looks professional and the "Rationale" sounds logical, how likely are you to stop "double-checking" the raw data after a 12-hour shift? How can we design the app to prevent this "lazy" trust?

- **Design Improvement:** Beyond the "Rationale Box," how else could you improve the design to reduce **cognitive load** or prevent **medical errors** in a high-pressure clinical environment?

Summary

You have evolved from a passive consumer of technology into an **AI-augmented clinician**. You are now capable of not only designing your own medical tools but—more importantly—critically auditing them to ensure they meet the highest standards of patient safety .